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Assignment-1 Sec:-10

CSE422

A) A* Search

A₁₀

A₁₀ B₁₄ F₉

F₉ B₁₄ G₇ H₁₃

G₇ B₁₄ H₁₃ I₈

I₈ B₁₄ H₁₃ H₁₂ J₁₀ E₁₅

J₁₀ B₁₄ H₁₃ H₁₂ E₁₅ E₁₅

A → F → G → I → J

Total cost = 10

G BF Search

A₁₀

A₁₀ B₈ F₆

F₆ B₈ G₅ H₃

H₃ B₈ G₅ I₁

I₁ B₈ G₅ J₀ E₃

J₀ B₈ G₅ E₃

A → F → H → I → J

Total cost = 15

B) A* Search

S₇

S₇ A₁₂ D₇

D₇ A₁₂ B₇ E₉

B₇ A₁₂ E₉ E₇ C₇

C₇ A₁₂ E₉ E₇ G₉

E₇ A₁₂ E₉ G₉ G₇

G₇ A₁₂ E₉ G₉

S → D → B → E → G

Total cost = 7

G BF Search

S₇

S₇ A₉ D₅

D₅ A₉ B₄ E₃

E₃ A₉ B₄ G₀

G₀ A₉ B₄

S → D → E → G

Total cost = 9

c)

1) Consistent heuristic value

$$h(a) = 0$$

$$h(b) = 0$$

$$h(c) = 0$$

$$h(d) = 0$$

$$h(e) = 0$$

$$h(f) = 0$$

$$h(g) = 0$$

$$h(h) = 0$$

$$h(i) = 0$$

2) Admissible heuristic value

$$h(a) = 0$$

$$h(b) = 0$$

$$h(c) = 0$$

$$h(d) = 0$$

$$h(e) = 0$$

$$h(f) = 0$$

$$h(g) = 0$$

$$h(h) = 0$$

$$h(i) = 0$$

3) Admissible and not Consistent heuristic value

$$h(a) = 24$$

$$h(b) = 0$$

$$h(c) = 0$$

$$h(d) = 0$$

$$h(e) = 0$$

$$h(f) = 0$$

$$h(g) = 0$$

$$h(h) = 0$$

$$h(i) = 0$$